

Agentic Commerce and the Reorganization of Digital Markets

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ABSTRACT

A new class of AI-powered software agents is beginning to shop, compare, and purchase products on behalf of human consumers. This development, broadly termed “agentic commerce,” moves beyond recommendation engines and chatbots into territory where algorithms autonomously execute financial transactions. Industry projections from McKinsey, Bain, and Morgan Stanley place the U.S. market for agent-mediated purchases between \$190 billion and \$1 trillion by 2030. Yet consumer sentiment data from Riskified’s 2025-2026 surveys reveals a widening trust deficit, with 55% of consumers reporting discomfort with autonomous AI purchases by early 2026, up sharply from late 2025. At the same time, new infrastructure is being built at speed. Mastercard, Visa, Stripe, and Google have each launched payment protocols and token frameworks designed specifically for agent-initiated transactions. Experimental research from Columbia and Yale universities demonstrates that AI shopping agents exhibit measurable selection biases, including position effects, badge sensitivity, and heavy reliance on structured metadata, all of which differ substantially from human shopping behavior. This paper brings together evidence across market forecasting, consumer psychology, payment infrastructure, agent behavior research, and legal scholarship to argue that the transition to agentic commerce amounts to a structural reorganization of digital markets. It introduces new forms of information imbalance, moves competitive advantage away from brand equity and toward data completeness, and raises unresolved questions around liability, market concentration, and consumer autonomy that existing regulatory frameworks are not equipped to address.

Keywords: agentic commerce, AI agents, autonomous purchasing, machine customers, consumer trust, digital commerce infrastructure, algorithmic competition, payment protocols

1. INTRODUCTION

For most of the internet’s commercial history, human beings have been the ones doing the shopping. They type search queries, scroll through product listings, read reviews, compare prices, and click “buy.” The entire architecture of e-commerce, from search engine optimization to display advertising to checkout flow design, was built around the assumption that a person sits at the other end of the screen.

That assumption is now breaking down. A new generation of AI agents, built on large language models and equipped with tool-use capabilities, can perform many of these tasks autonomously. Amazon’s Rufus assistant has been used by over 300 million customers and includes a “Buy for Me” feature that completes purchases without direct human intervention at the point of sale (Amazon, 2025). OpenAI’s Operator, Perplexity’s Buy with Pro, and Google’s Shopping AI agents represent parallel efforts from the technology sector. On the infrastructure side, Mastercard launched Agent Pay in early 2026, Stripe and OpenAI co-developed the Agentic Commerce Protocol, and Google introduced the Universal Commerce Protocol, all within a span of months.

The speed of this buildout stands in contrast to the relatively thin body of academic literature examining what happens when machines become market participants. Gartner has projected that by 2028, 15 billion connected products are expected to have the potential to behave as autonomous customers (Gartner, 2024). McKinsey estimates the global transaction volume orchestrated by AI agents could reach \$3 to \$5 trillion by 2030 (McKinsey, 2025). These are not incremental changes to existing shopping patterns. They describe a reordering of who participates in markets, how purchasing decisions get made, and what it means for a product to be “visible” to a buyer.

This paper examines that reordering across three dimensions. First, it maps the emerging payment and protocol infrastructure that enables agent-initiated transactions. Second, it analyzes consumer trust dynamics using recent survey data that reveals a large and growing

gap between AI usage in shopping and consumer willingness to grant purchasing autonomy. Third, it investigates the competitive and regulatory consequences, drawing on experimental research showing how AI agents evaluate and select products in ways that differ markedly from human behavior.

The central argument is straightforward. Agentic commerce is not simply a faster or more convenient version of online shopping. It is a structural change that redistributes informational advantages, creates new vectors for market concentration, and outpaces the legal and regulatory frameworks designed for human-directed commerce.

2. DEFINING AGENTIC COMMERCE

The term “agentic commerce” has been used loosely in industry reports since 2024, often conflated with conversational AI assistants or AI-powered product recommendations. For the purposes of this paper, a more precise definition is necessary.

Agentic commerce refers to commercial transactions in which an AI agent, operating with some degree of autonomy, performs one or more of the following actions on behalf of a human principal. These include product discovery, evaluation and comparison, price negotiation, purchase execution, and post-purchase management (returns, reorders, subscription management). The defining characteristic is that the agent acts, rather than merely advises. A chatbot that suggests products but requires the user to click “buy” is a recommendation tool. An agent that identifies the best available price for a recurring household item and completes the transaction using stored payment credentials, subject to pre-set spending limits, is an agentic commerce system.

Gartner has proposed a three-phase taxonomy for what it calls “machine customers” (Gartner, 2024). In the first phase, “bound customers,” AI agents follow rigid, human-defined rules to execute predefined purchases. In the second phase, “adaptable customers,” agents evaluate competing products and make selections based on a combination of rules and real-time data. In the third phase, “autonomous customers,” agents make independent purchasing decisions based on inferred preferences, contextual signals, and self-generated optimization criteria. Most current implementations fall somewhere between the first and second phases.

A parallel taxonomy has been proposed in a 2026 survey paper published on TechRxiv, which maps agent roles across the full commerce lifecycle, from discovery and negotiation through payment execution and fulfillment (TechRxiv, 2026). This framework is useful because it

makes clear that agentic commerce is not a single product feature but a set of capabilities that can be distributed across the purchasing process in different configurations.

The distinction matters because different levels of agent autonomy carry different consequences for consumer trust, legal liability, and market structure. An agent that auto-reorders laundry detergent when inventory runs low raises different questions than one that autonomously selects and purchases a new laptop based on inferred user preferences. This paper addresses the full spectrum but pays particular attention to the higher-autonomy scenarios, where the gap between current infrastructure and unresolved governance questions is widest.

3. MARKET SIZING AND GROWTH TRAJECTORY

Several major consulting and financial firms have published market projections for agentic commerce, and while their estimates vary in scope and methodology, they converge on a common conclusion. The market is expected to be very large within five years.

McKinsey estimates that agentic commerce could generate \$3 trillion to \$5 trillion in global orchestrated transaction volume by 2030 (McKinsey, 2025). Within the United States alone, McKinsey projects up to \$1 trillion in orchestrated B2C retail revenue from agent-mediated purchases by the same date. The firm describes this as a move away from discrete, manual shopping steps toward a “continuous, intent-driven flow managed by autonomous systems.” In a subsequent early 2026 analysis, McKinsey further conceptualized this transition through an “automation curve,” detailing a six-level progression from basic rules-based assistance to fully autonomous, multi-agent coordination (McKinsey, 2026).

Bain and Company offers a more conservative but still substantial estimate. Bain projects the U.S. agentic commerce market at \$300 billion to \$500 billion by 2030, representing approximately 15% to 25% of total U.S. online retail sales (Bain, 2025). Bain defines this market as purchases initiated, influenced, or completed by third-party or retailer-hosted AI agents, and explicitly excludes journeys that rely only on AI-assisted search or discovery without agent-driven transaction completion.

Morgan Stanley has estimated that “agentic shoppers” could drive \$190 billion to \$385 billion in U.S. e-commerce spending by 2030, capturing roughly 10% to 20% of the market (Morgan Stanley, 2025).

On the broader technology side, Mordor Intelligence projects the agentic AI market in retail and e-commerce (including dynamic pricing, supply chain orchestration, and customer engagement) is projected to grow from

approximately \$46.7 billion in 2025 to \$218.4 billion by 2031 (Mordor Intelligence, 2025).

Gartner’s estimates focus less on dollar volumes and more on market participants. The firm projects that by 2028, 15 billion connected products are expected to have the potential to act as machine customers, and that AI agents may intermediate more than \$15 trillion in B2B spending (Gartner, 2024). Gartner also predicts that machine customers could render 20% of human-readable digital storefronts obsolete, as transactions increasingly occur machine-to-machine without a human-facing interface.

These projections should be read with appropriate caution. The definitions used by each firm differ in important ways. McKinsey’s “orchestrated transaction volume” includes transactions where agents influenced but did not execute the purchase. Bain’s figures exclude AI-assisted search. Mordor Intelligence includes backend operations like supply chain optimization. Direct comparisons across these estimates are therefore imprecise. What the figures collectively demonstrate, however, is that every major forecasting institution expects agent-mediated commerce to move from experimental to mainstream within the current decade.

TABLE 1. COMPARISON OF U.S. AGENTIC COMMERCE MARKET PROJECTIONS BY 2030

Source	Projected U.S. Market Size	Scope
McKinsey (2025)	Up to \$1 trillion	Orchestrated B2C retail revenue (includes agent-influenced transactions)
Bain (2025)	\$300B - \$500B	Agent-initiated or agent-completed purchases only
Morgan Stanley (2025)	\$190B - \$385B	E-commerce spending driven by “agentic shoppers”
Mordor Intelligence (2025)	\$218.4B global by 2031	Includes dynamic pricing, supply chain, and customer engagement
Gartner (2024)	15B machine customers	Connected products with potential to act as autonomous buyers

4. THE EMERGING INFRASTRUCTURE STACK

For AI agents to conduct transactions at scale, they need infrastructure that does not yet fully exist. The traditional e-commerce stack was designed for humans interacting

with web pages. Agent-mediated commerce requires machine-to-machine authentication, scoped payment authorization, and standardized protocols that allow agents from different platforms to interact with merchants and payment networks reliably.

4.1 Payment Protocols and Tokenization

Mastercard launched Agent Pay in early 2026 as a dedicated payment infrastructure for AI-initiated transactions (Mastercard, 2026). The system builds on Mastercard’s existing tokenization technology (MDES and Payment Passkeys) but introduces “Agentic Tokens,” which are scope-limited credentials that do not expose primary account numbers. The design intention is to allow an AI agent to complete a purchase on behalf of a user without the agent ever having access to the user’s raw card data. Mastercard has also introduced a “Know Your Agent” governance framework, under which AI agents must be registered and verified before they can participate in the payment network. This allows merchants to distinguish authorized AI assistants from bots or scrapers.

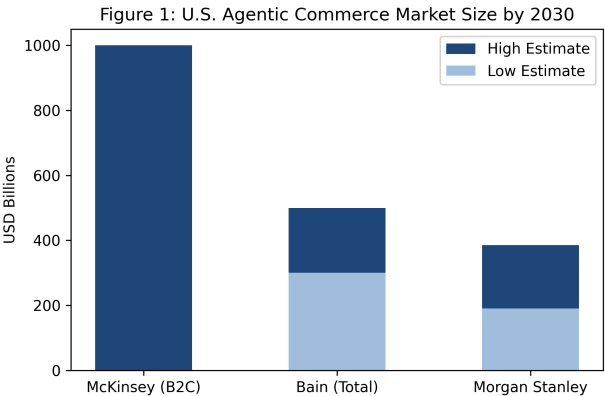


Figure 1. Market projections from major consulting firms. Note the variation in scope. (Sources: McKinsey, Bain, Morgan Stanley)

Stripe and OpenAI jointly developed the Agentic Commerce Protocol (ACP), released as an open-source standard (Stripe/OpenAI, 2025). ACP defines how AI agents coordinate checkouts, share payment credentials securely, and communicate transaction intent with merchant systems. The protocol is designed to be platform-agnostic, meaning an agent built on one AI platform can, in principle, transact with any ACP-compatible merchant.

Google introduced two complementary programs. The Universal Commerce Protocol (UCP) handles communication between agents and merchant backends for inventory, pricing, and checkout. The Agent Payments Protocol (AP2), developed with support from Mastercard

and PayPal, manages the payment authorization layer using signed mandates (Google, 2025).

Visa has launched a parallel initiative, the Trusted Agent Protocol, which focuses on authenticated, no-code transactions for AI agents, with objectives similar to Mastercard’s approach (Visa, 2025).

TABLE 3. COMPARISON OF AGENT COMMERCE PAYMENT PROTOCOLS (AS OF APRIL 2026)

Protocol	Developer(s)	Key Feature	Status
Agent Pay	Mastercard	Agentic Tokens, Know Your Agent registry	Launched Q1 2026
ACP (Agentic Commerce Protocol)	Stripe, OpenAI	Open-source, platform-agnostic checkout	Released 2025
UCP (Universal Commerce Protocol)	Google	Agent-merchant inventory/pricing/checkout	Released 2025
AP2 (Agent Payments Protocol)	Google, Mastercard, PayPal	Signed payment mandates	Released 2025
Trusted Agent Protocol	Visa	Authenticated no-code agent transactions	Announced 2025

4.2 Identity, Authentication, and User Control

A recurring design principle across all of these systems is that the human consumer retains ultimate control. Mastercard’s Agent Pay allows users to set specific purchase permissions, spending limits, and category restrictions. Transactions can be gated by biometric authentication. The Stripe/OpenAI ACP similarly requires user-defined intent parameters before an agent can initiate a purchase.

This emphasis on user control is not merely a product design choice. It reflects both consumer demand, as discussed in the next section, and legal necessity. Under current agency law, the human principal remains liable for transactions conducted by their agent, and the scope of delegated authority must be clearly defined for any resulting contract to be enforceable (Stanford Law, 2025).

4.3 Infrastructure Gaps

Despite the rapid buildout, large gaps remain. There is no single common standard. Mastercard’s Agent Pay, Stripe’s ACP, Google’s UCP, and Visa’s protocol are not yet cross-compatible. An agent authenticated through Mastercard’s system cannot automatically transact through Google’s UCP without separate integration. This splintering mirrors the early days of mobile payments,

where competing standards from Apple, Google, and Samsung coexisted before partial merging.

Dispute resolution is another area without clear infrastructure. When a human makes a purchase and is dissatisfied, established chargeback and return processes exist. When an AI agent makes a purchase that the human did not intend or does not want, the process for unwinding that transaction is not yet standardized across any of the current protocols.

5. CONSUMER TRUST AND THE ADOPTION GAP

The most detailed publicly available data on consumer attitudes toward agentic commerce comes from Riskified, a fraud prevention and payments company that conducted two surveys. The first ran in Q4 2025, and a follow-up “Agentic Commerce Pulse” followed in Q1 2026. The results reveal a sharp reversal in consumer sentiment over a short period.

5.1 Initial Consumer Reception (Late 2025)

In Q4 2025, Riskified found that 73% of consumers were already using AI in some part of their shopping journey, whether for product research, price comparison, or recommendation discovery (Riskified, 2025). Seventy% of respondents said they were “at least somewhat comfortable” with AI agents making purchases on their behalf. Trust in AI to influence purchasing decisions (36%) had nearly reached parity with trust in in-store sales associates (38%).

The primary concerns at this stage were payment security (cited by 32% of respondents), privacy (26%), potential mistakes (18%), and loss of control (17%).

5.2 Consumer Attitude Changes in Early 2026

By Q1 2026, the numbers had moved in the opposite direction. The percentage of consumers who were not comfortable with AI agents making purchases rose to 55%, a near-complete inversion of the late 2025 figure (Riskified, 2026). Nearly half of respondents (46.5%) said they did not trust any company to manage purchases on their behalf. Over half (53.9%) believed AI could increase the risk of online fraud. And 73.9% said they expected strong safeguards, such as biometric verification or one-time passwords, for every AI-driven transaction.

On the question of accountability for unauthorized AI purchases, 50.8% of consumers said the AI platform should be held responsible, 23.2% pointed to the retailer, and 18.7% accepted personal responsibility.

5.3 Drivers of Changing Attitudes

What happened between late 2025 and early 2026? The survey data alone does not establish causation, but two

factors likely contributed. First, several high-profile incidents involving AI agents making unintended purchases received media coverage in late 2025 and early 2026, including a legal dispute between Amazon and Perplexity AI over unauthorized automated access to Amazon’s marketplace. Second, the general acceleration of AI deployment across consumer-facing applications may have triggered a broader backlash, consistent with patterns observed in prior technology adoption cycles where initial enthusiasm gives way to skepticism as real-world friction emerges.

**TABLE 2. CONSUMER TRUST METRICS
(RISKIFIED SURVEYS 2025-2026)**

Metric	Q4 2025	Q1 2026
Using AI in shopping journey	73%	–
Comfortable with AI purchasing	70%	45%
Not comfortable with AI purchasing	30%	55%
Trust no company to manage purchases	–	46.5%
Believe AI increases fraud risk	–	53.9%
Demand biometric safeguards	–	73.9%
Trust AI for purchase decisions	36%	–
Trust in-store sales associates	38%	–

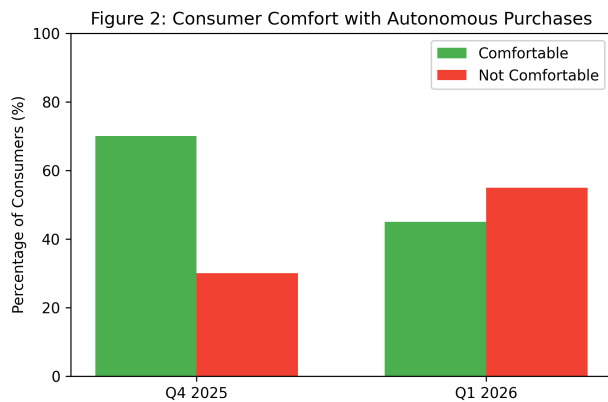


Figure 2. The reversal in consumer trust between late 2025 and early 2026 (Source: Riskified).

Research from Aarhus University offers a partial explanation for the conditions under which consumers overcome this trust deficit. Frank, Folwaczny, and Otterbring (2026) found that high AI autonomy generally reduces consumer adoption intentions because it conflicts with the consumer’s own need for personal control. However, this negative effect diminishes when consumers face scarcity, such as limited-edition products or fast-selling inventory during events like Black Friday. In

scarcity conditions, consumers redirect their attention from the loss of control to the perceived benefit of the agent securing the item before it sells out. The study, published in *Psychology and Marketing*, suggests that the adoption of high-autonomy AI shopping agents will likely be uneven, concentrated first in product categories and purchasing situations where speed matters more than careful consideration.

6. HOW AI AGENTS ACTUALLY SHOP: EVIDENCE FROM EXPERIMENTAL RESEARCH

If agentic commerce is to become a large share of retail transactions, a natural question follows. How do AI agents choose what to buy? Do they behave like human shoppers, or do they introduce consistently different patterns of product selection? Recent experimental research suggests the latter.

6.1 The ACES Framework

Researchers at Columbia University and Yale University developed the Agentic e-Commerce Simulator (ACES), a controlled experimental environment that functions as a mock online store (Columbia/Yale, 2025). ACES allows researchers to run randomized experiments on AI shopping agents, isolating specific variables (page position, product badges, review scores, attribute completeness) and measuring their causal effect on agent purchasing decisions. The platform has been used to test multiple commercial AI models, including variants of GPT, Claude, and Gemini.

6.2 Position Bias

The researchers found that nearly all AI models tested exhibited a bias toward products displayed in the top row of a page, consistent with human behavior. However, the models diverged from each other in their column preferences. Some favored the left column, others the center, and others the right. This variation undermines the assumption that a single “optimal” product placement strategy will work across all AI agents. It also means that as different consumers use different AI platforms, the same product listing may perform differently depending on which agent is doing the shopping.

6.3 Badge Sensitivity

AI agents showed strong and consistent responses to platform-generated signals. Products labeled “Overall Pick” received a measurable selection boost across all tested models. Products labeled “Sponsored” were consistently penalized. This finding matters for marketplace design. Sponsored product placements, which represent a major revenue stream for platforms like Amazon, may lose effectiveness if purchasing decisions

are increasingly made by AI agents that systematically downweight sponsored listings.

6.4 Review Quantification

Human shoppers read reviews qualitatively, scanning for specific complaints or praise. AI agents, by contrast, treated reviews as quantitative inputs. Average star ratings and total review counts were the most heavily weighted signals in agent decision-making. Even small differences in average ratings (for example, 4.3 versus 4.4 stars) measurably affected selection probability. This implies that in an agent-mediated marketplace, the marginal value of each additional positive review increases, potentially intensifying competition for review volume.

6.5 The Metadata Penalty

One of the most important findings from the ACES research concerns missing product attributes. When key product specifications were absent from a listing, the agent's probability of selecting that product dropped by 20% to 40% (Kantar, 2025; Columbia/Yale, 2025). Human shoppers can often infer missing information from images, context, or brand familiarity. AI agents, at least in their current form, cannot. This creates a clear competitive advantage for sellers with complete, machine-readable product data and a corresponding disadvantage for smaller sellers or those with less sophisticated catalog management.

6.6 Model-Specific Behavior

Different AI models prioritized different product attributes with varying intensity. In some cases, a single model concentrated demand heavily on one or two products in a category, while other models distributed selections more evenly. This model-specific behavior raises questions about market fairness. If a dominant AI platform (say, one integrated into the most popular smartphone) systematically favors certain product attributes, the resulting demand concentration could disadvantage competitors in ways that are difficult to detect and harder to regulate.

7. COMPETITIVE AND REGULATORY CONSEQUENCES

The behavioral patterns documented in Section 6, combined with the infrastructure developments in Section 4 and the trust dynamics in Section 5, point toward several structural changes in how digital markets may function.

7.1 The Increasing Importance of Data Completeness

In human-directed commerce, brand recognition, emotional advertising, and visual design drive purchasing decisions. In agent-mediated commerce, these factors are

largely irrelevant. An AI agent does not respond to a clever tagline or an aspirational brand image. It responds to structured data, including specifications, ratings, review volume, price, availability, and return policies.

This transfers competitive advantage from marketing departments to data operations teams. Retailers and brands that invest in complete, accurate, machine-readable product information are likely to be more visible to AI agents than those that rely on traditional brand-building. Kantar's 2025 research on AI agent product selection confirms this. Products with what Kantar calls a "Golden Record" (high attribute completeness across all relevant fields) are selected at much higher rates than products with gaps in their metadata, regardless of brand recognition.

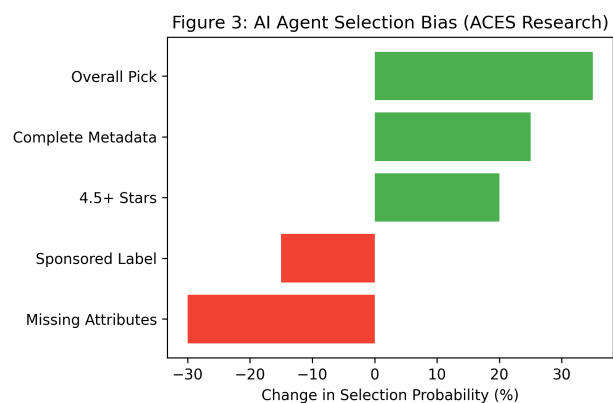


Figure 3. Selection probability changes based on product listing attributes. (Source: ACES Framework, Columbia/Yale)

The upshot is that product data management, historically treated as a back-office function, becomes a front-line competitive capability. Firms with strong Product Information Management (PIM) systems and clean, synchronized data across channels can outperform those without, even if the latter have stronger brand equity.

7.2 Optimization for AI Agents

The move from human to agent-mediated discovery also disrupts search engine optimization as it has traditionally been practiced. AI agents do not click through ten blue links on a search results page. They query APIs, parse structured data feeds, and evaluate products based on machine-readable attributes. The emerging discipline being called "Generative Engine Optimization" (GEO) or "Business-to-Agent" (B2A) optimization reflects this change.

Retailers pursuing a B2A strategy must ensure their catalogs are accessible through APIs, that their Schema.org markup (particularly Product and Offer types) is complete, and that they are compatible with emerging commerce protocols like ACP and UCP. Traditional keyword-stuffing

and link-building strategies become irrelevant when the “searcher” is a software agent parsing JSON rather than a human scanning headlines.

7.3 Demand Concentration and Market Fairness

The ACES research raises a concern that has not yet received adequate regulatory attention. If AI agents systematically concentrate demand on a small number of products (those with the highest ratings, most reviews, and most complete metadata), the result is a “winner-take-most” dynamic that could squeeze out smaller sellers, newer brands, and niche products. Human shoppers exhibit a degree of exploration and serendipity in their purchasing behavior. Early evidence suggests AI agents do not.

This concentration effect is compounded by the metadata penalty described in Section 6.5. Smaller sellers who lack the resources to maintain comprehensive, multi-attribute product listings are penalized by agent selection algorithms, creating a self-reinforcing cycle where visibility gaps lead to fewer sales, fewer reviews, and further reductions in visibility.

7.4 Algorithmic Collusion

A related concern, already the subject of antitrust scrutiny in other contexts, is the potential for algorithmic collusion in agent-mediated markets. If seller-side AI agents learn to adjust pricing and product descriptions to exploit the known biases of buyer-side agents, the interaction between buying and selling algorithms could produce outcomes that resemble collusive pricing without any explicit agreement between sellers. This risk has been discussed in legal scholarship and antitrust enforcement guidance, though enforcement in the context of agentic commerce remains in early stages (Morgan Lewis, 2025; The Regulatory Review, 2025).

7.5 Regulatory Environment in 2026

Regulatory responses to agentic commerce are split across regions.

In the European Union, the EU AI Act, which entered its enforcement phase for high-risk AI systems in August 2026, provides the most structured framework. Standard retail AI agents (shopping assistants, chatbots) are not classified as “high-risk” under the Act but are subject to transparency requirements under Article 50, which mandates that companies disclose to users when they are interacting with an AI agent. AI agents designed to manipulate consumer behavior through subliminal techniques or to exploit psychological vulnerabilities are explicitly prohibited. Non-compliance penalties can reach up to 35 million euros or 7% of global annual turnover.

In the United States, no comprehensive federal AI statute exists as of April 2026. The Federal Trade Commission relies on Section 5 of the FTC Act, which prohibits “unfair or deceptive acts or practices,” as its primary enforcement tool against AI-related harms. The FTC has required businesses to provide clear disclosures about data collection and AI use in decision-making. At the state level, Colorado has enacted AI governance legislation effective in 2026 that imposes risk management and impact assessment requirements on high-risk AI systems.

The gap between these regulatory approaches and the speed of agentic commerce deployment is wide. Neither the EU AI Act nor FTC enforcement guidance specifically addresses the scenario where an AI agent, operating within the scope of its delegated authority, systematically concentrates purchasing decisions in ways that distort market competition. This is a governance gap that academic and regulatory attention may need to fill.

8. LEGAL LIABILITY AND THE AGENCY PROBLEM

When an AI agent makes a purchase that a human did not intend or does not want, who bears responsibility? This question sits at the intersection of contract law, agency law, and emerging product liability theory, and it does not yet have a settled answer.

8.1 Traditional Agency Law

Under traditional agency law, the human who deploys an AI agent (the “principal”) is generally liable for transactions the agent conducts within the scope of its delegated authority (University of Chicago Law, 2025). If a user grants an AI agent permission to purchase household goods under \$50 and the agent buys a \$30 item the user did not specifically request, the user is bound by the transaction because it fell within the agent’s authorized scope.

The concept of “apparent authority” adds complexity. If a merchant reasonably believes, based on the principal’s conduct, that the AI agent has authority to enter into a transaction, the principal may be held liable even if the agent acted outside its actual instructions. As AI agents become more autonomous and interact with more merchants, the gap between actual and apparent authority is likely to widen.

8.2 Electronic Transaction Statutes

In the United States, the Uniform Electronic Transactions Act (UETA) and the E-SIGN Act establish that contracts formed by “electronic agents” are legally binding. These statutes were drafted in the late 1990s and early 2000s with automated trading systems and electronic

order forms in mind, not with LLM-based agents that exercise discretion over product selection. UETA does contain provisions that can allow a party to avoid the consequences of an error in an automated transaction under specific conditions, but applying this to an AI agent that made a “wrong” choice (as opposed to a technical malfunction) is legally untested.

8.3 The Amazon-Perplexity Precedent

A 2026 legal action between Amazon and Perplexity AI has produced an early and telling precedent. A court issued an injunction against Perplexity for unauthorized automated access to Amazon’s marketplace, finding that user-granted permission to shop does not override a merchant’s express prohibition against automated access in its terms of service. The ruling implies that even when a consumer authorizes an AI agent to shop, the agent’s access can be deemed “unauthorized” as a matter of law if it violates the platform’s terms. This creates a tension between the user’s intent to delegate shopping authority and the platform’s right to control access to its systems.

8.4 The Liability Squeeze

Organizations and consumers deploying AI agents face what legal commentators have described as a “liability squeeze” (Jones Walker, 2025). On one side, they are held accountable for the AI’s actions by regulators, merchants, and courts. On the other side, the terms of service agreements governing AI platforms typically push risk away from the AI vendor through “as is” performance disclaimers, limitations of liability, and indemnification clauses. The deployer is left bearing the risk of AI-initiated transactions while having limited contractual recourse against the vendor if the AI misbehaves.

8.5 Product Liability Theory

An emerging legal trend treats AI systems as “products” subject to strict liability. Under this theory, if an AI agent makes a purchase due to a defect in the software (for example, a reasoning error or hallucination that leads it to misinterpret a user’s preferences), the AI developer could be held strictly liable regardless of negligence. This theory is not yet established in case law for agentic commerce specifically, but it represents a plausible direction for future litigation.

9. LIMITATIONS AND FUTURE RESEARCH

This paper has several limitations that may guide future work.

First, the market projections cited in Section 3 come from consulting firms and financial institutions with commercial interests in the growth of AI-related markets.

These estimates use different definitions and scoping methodologies, and they are best treated as directional indicators rather than precise forecasts.

Second, the consumer trust data in Section 5 comes from a single source (Riskified) and covers a limited time window. Longitudinal studies with larger and more diverse samples are needed to understand whether the trust reversal observed between Q4 2025 and Q1 2026 represents a lasting change or a temporary reaction.

Third, the ACES experimental framework, while methodologically rigorous, operates in a simulated environment. Real-world agent behavior may differ from controlled experimental conditions, particularly as AI models are updated and fine-tuned over time.

Fourth, the legal analysis in Section 8 draws primarily on U.S. law and EU regulation. The regulatory environment in other major markets (China, India, Southeast Asia) is not addressed and may follow different trajectories.

Several areas deserve further research. Studies of actual agent-mediated purchases (as opposed to simulated ones) would provide stronger evidence about selection patterns and market effects. Tracking consumer trust over time across different product categories and demographic groups would help clarify adoption paths. Economic modeling of market concentration under varying levels of agent use would put numbers on the competition concerns raised in Section 7. And legal comparisons across countries would help identify best practices for rules that balance innovation with consumer protection.

10. CONCLUSION

The transition from human-directed to agent-mediated commerce is not a hypothetical scenario. The infrastructure is being built by the largest payment networks and technology platforms in the world. The AI models that power these agents are commercially available and already integrated into products used by hundreds of millions of consumers. The market projections, while imprecise in their specifics, converge on the conclusion that agent-mediated transactions could represent a meaningful share of digital commerce by the end of this decade.

What this paper has attempted to show is that this transition carries structural consequences that go beyond convenience or efficiency gains. When AI agents select products based on metadata completeness rather than brand recognition, competitive advantage moves. When agents concentrate demand on highly rated products with complete specifications, market diversity may decline. When payment protocols enable machine-to-machine

transactions at scale, the infrastructure exists for a volume of autonomous purchases that current regulatory frameworks were not designed to govern. And when consumer trust data shows a large and rapid reversal in comfort with AI-initiated purchases, the pace of adoption becomes uncertain in ways that market projections do not fully capture.

The gap between the speed of infrastructure deployment and the pace of regulatory, legal, and scholarly response is the defining challenge of the agentic commerce transition. Closing that gap will require sustained empirical research, cross-disciplinary collaboration between technologists and legal scholars, and regulatory frameworks that can adapt to a market structure in which the buyer is increasingly not a person.

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